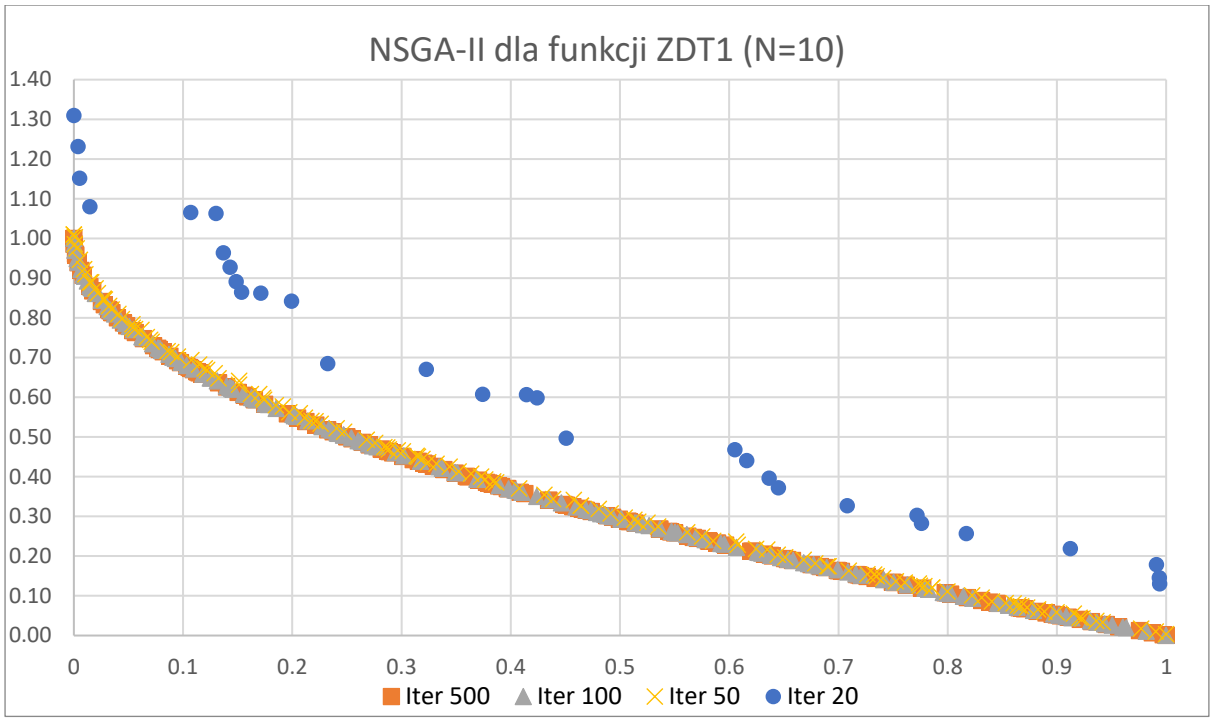
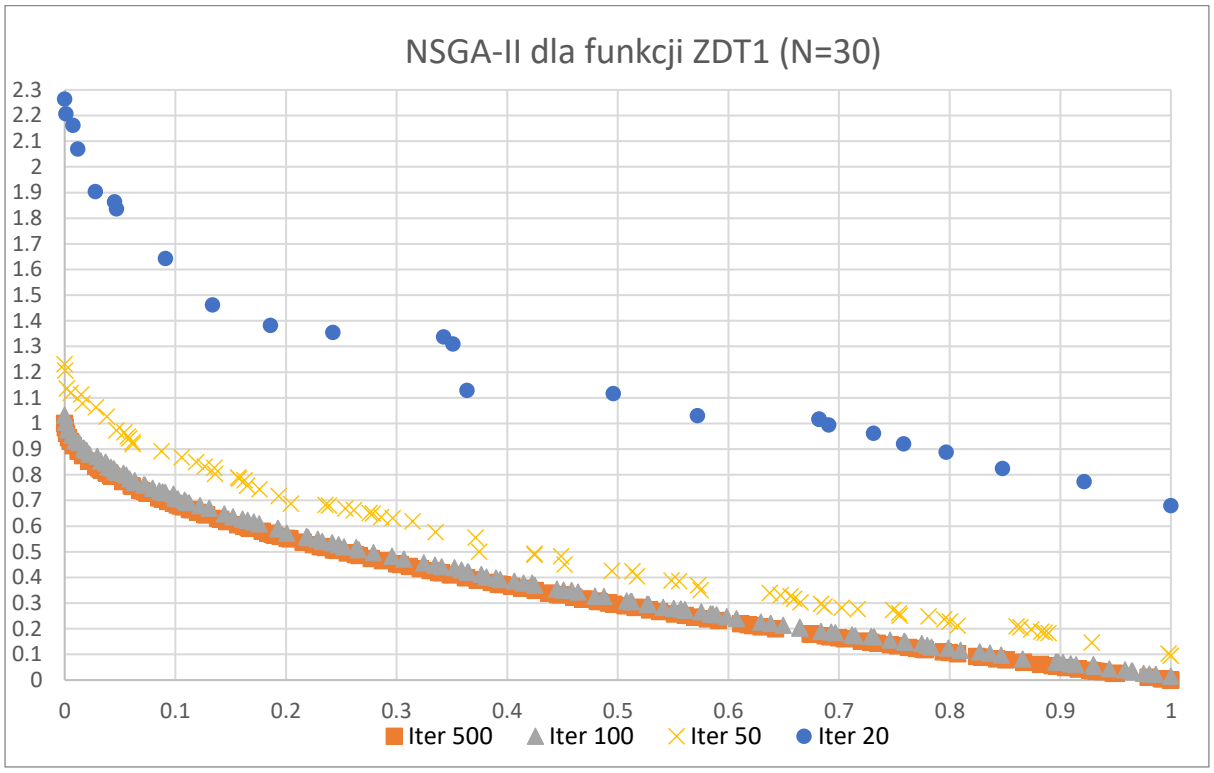
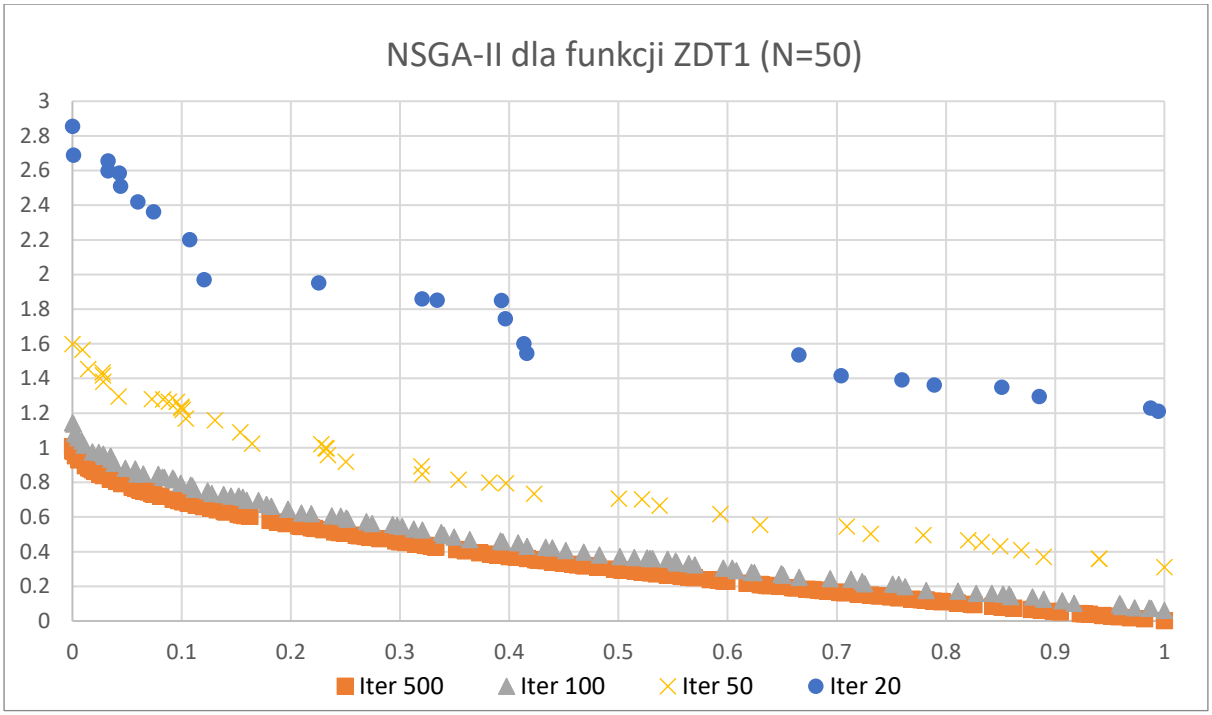
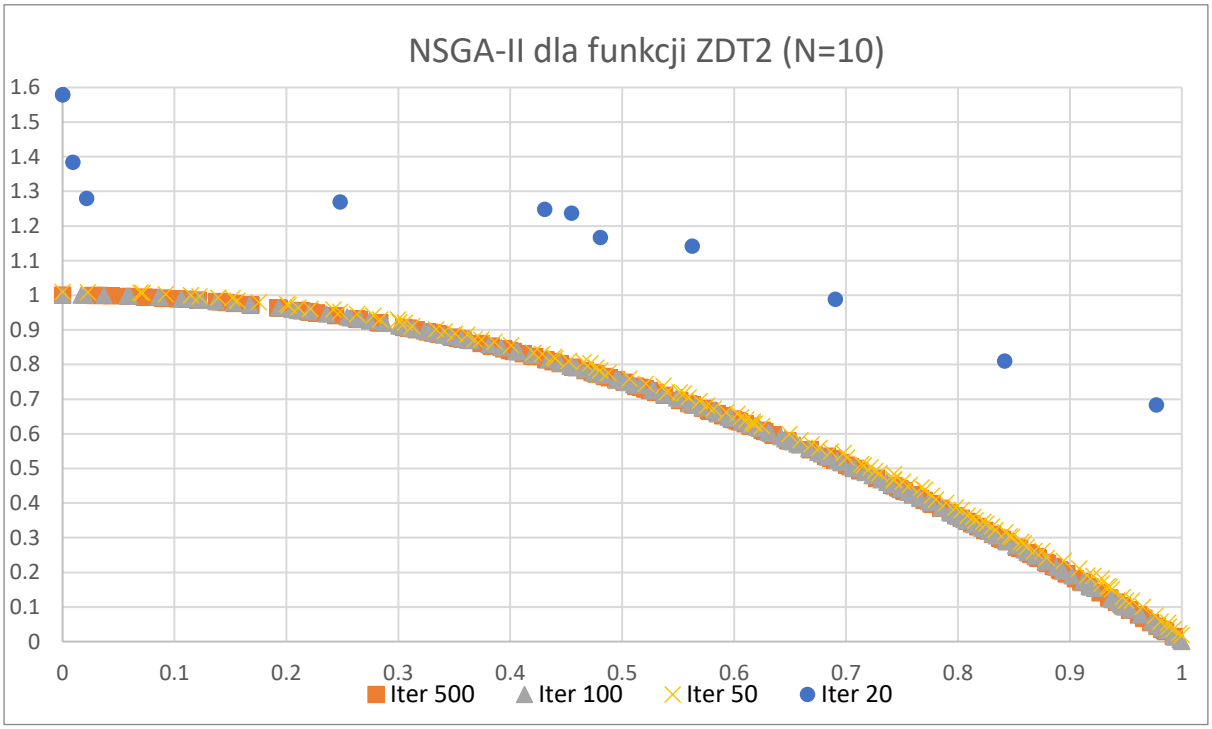
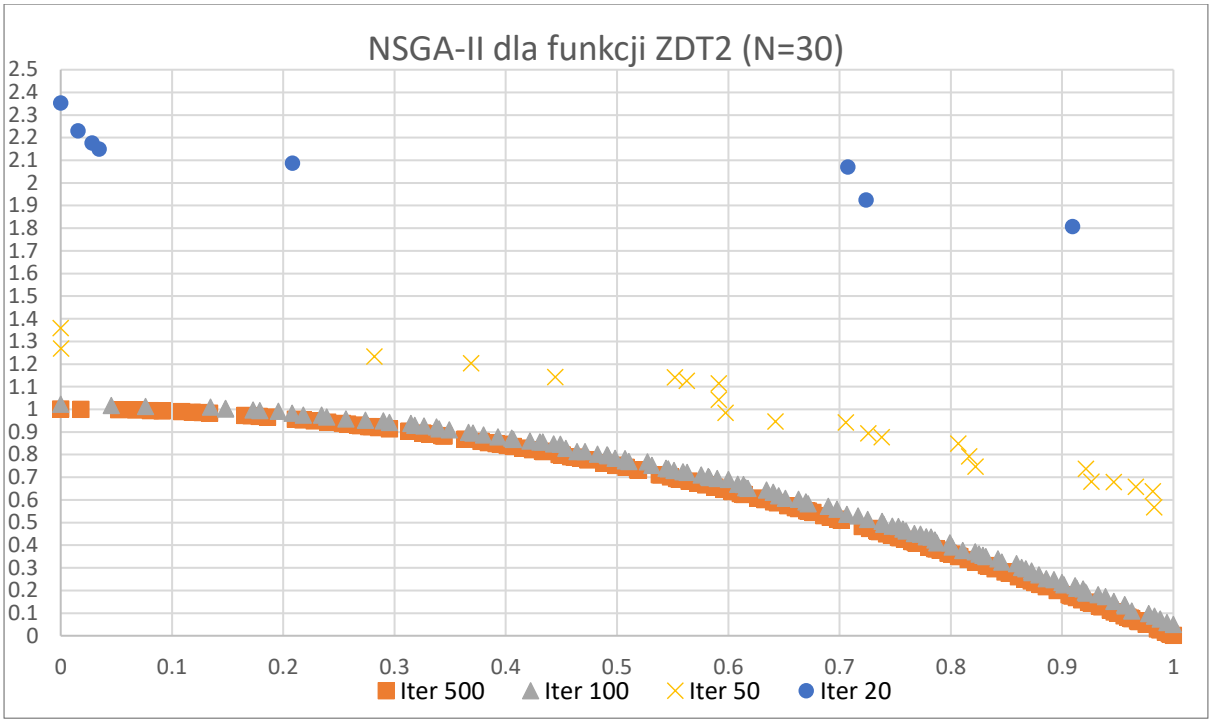
Wyniki algorytmu dla funkcji ZDT1-4 + ZDT6:

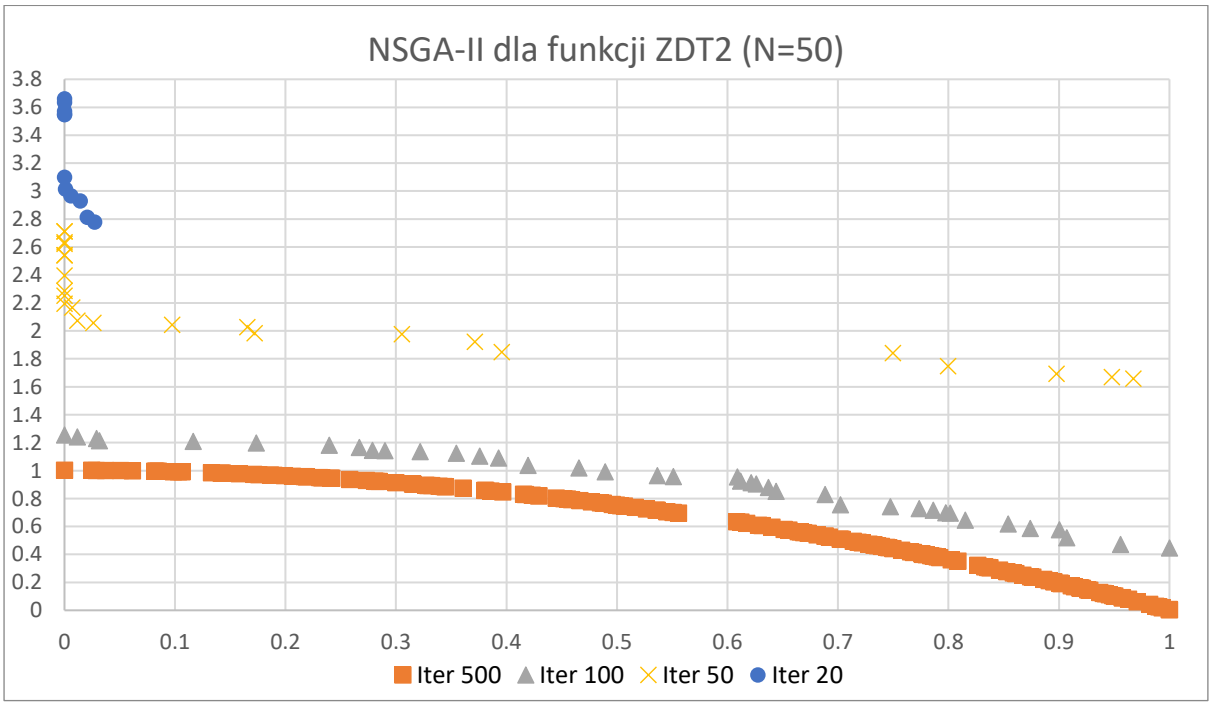


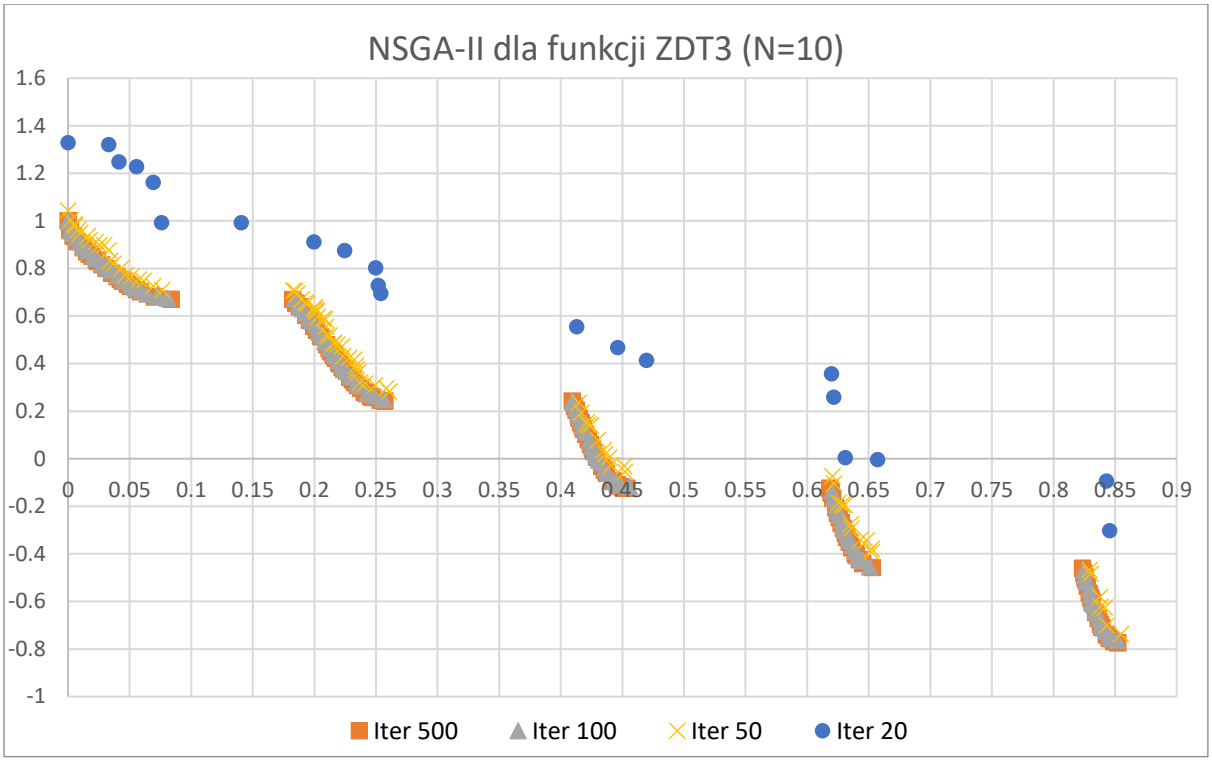


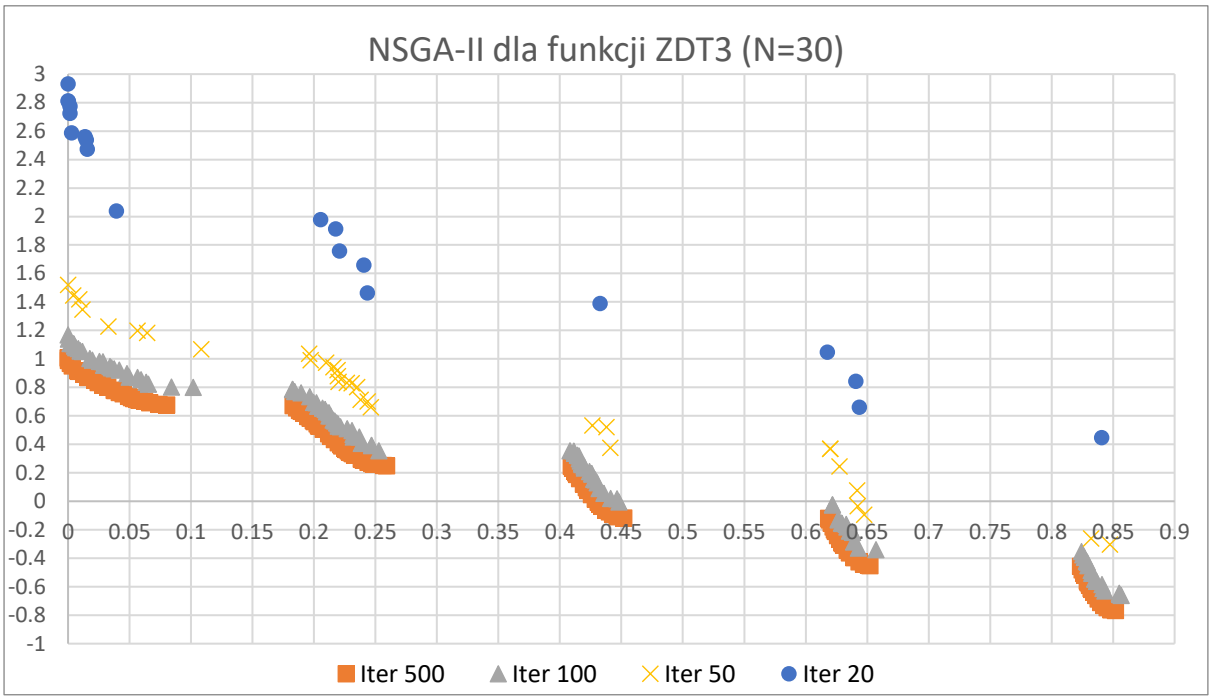

NSGA-II dla funkcji ZDT1 (N=50)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Iter 500
Iter 100
Iter 50
Iter 20

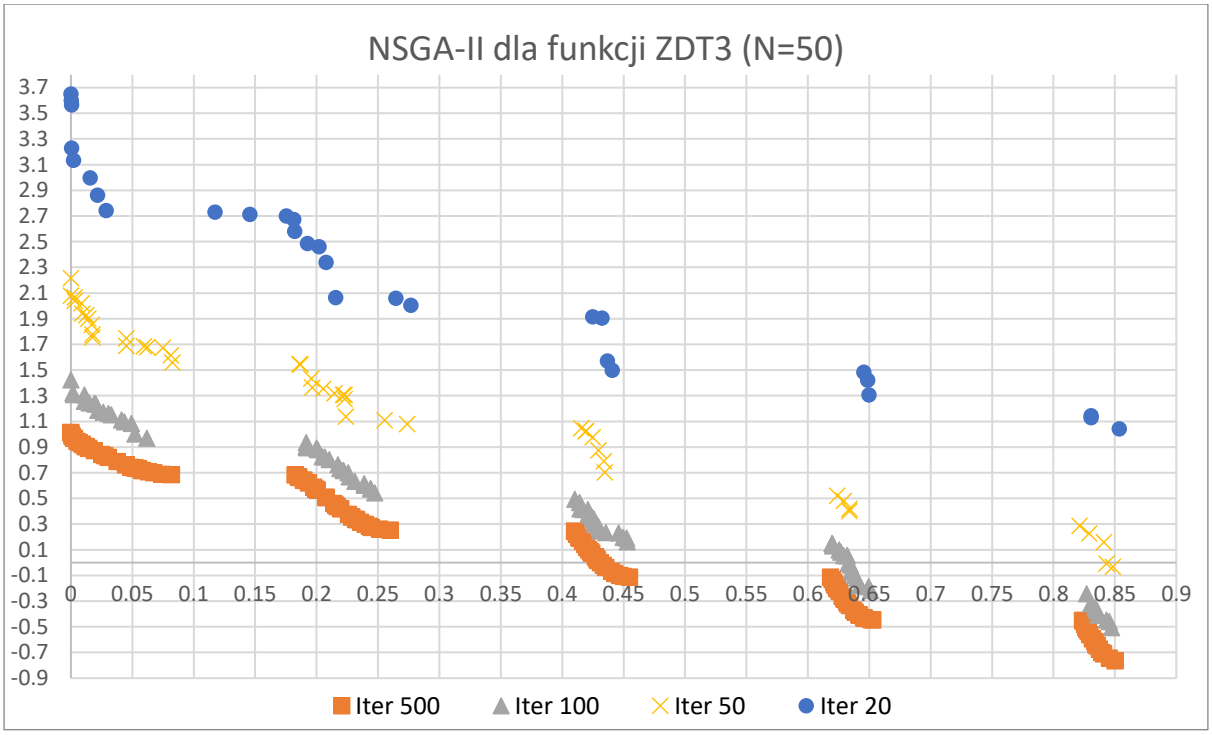

NSGA-II dla funkcji ZDT2 (N=10)
1.6
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Iter 500
Iter 100
Iter 50
Iter 20

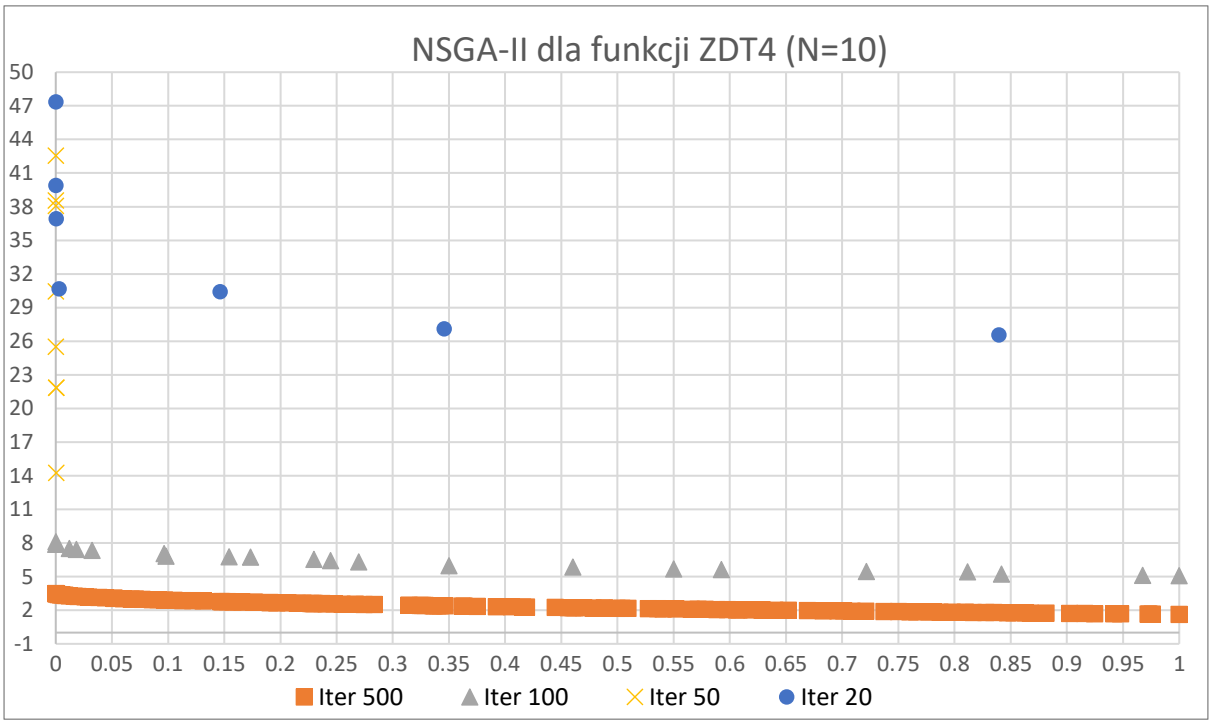

NSGA-II dla funkcji ZDT2 (N=30)
2.5
2.4
2.3
2.2
2.1
2
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Iter 500
Iter 100
Iter 50
Iter 20

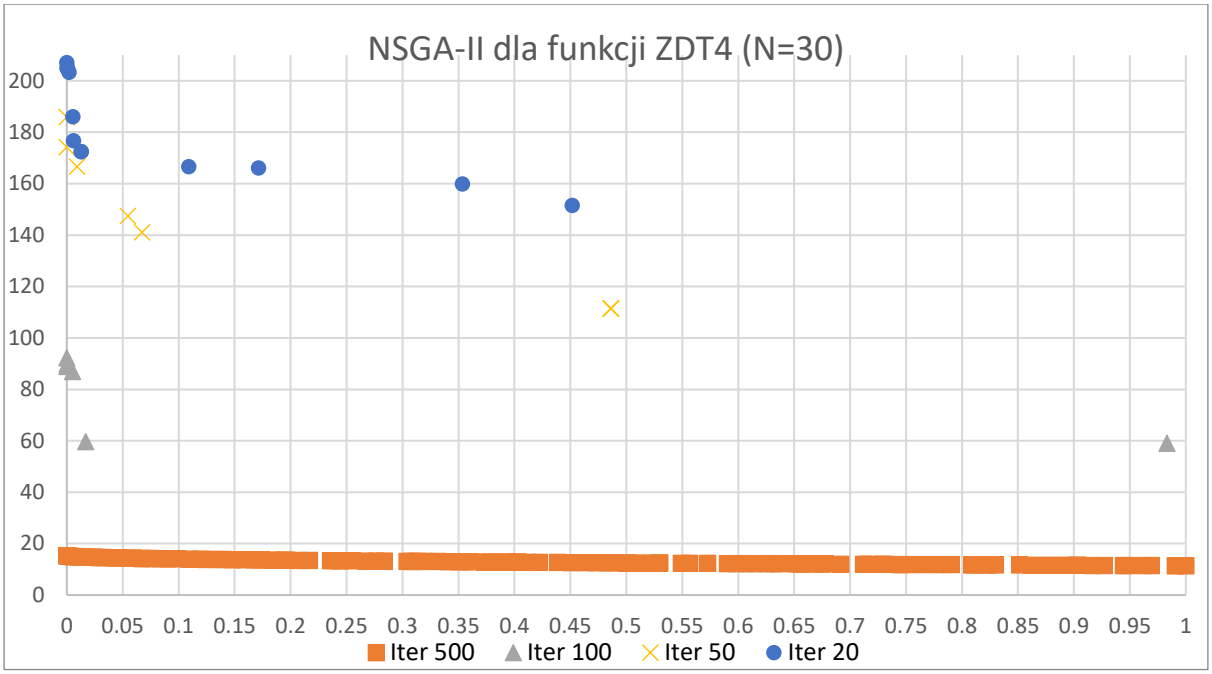

NSGA-II dla funkcji ZDT2 (N=50)
3.8
3.6
3.4
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Iter 500
Iter 100
Iter 50
Iter 20

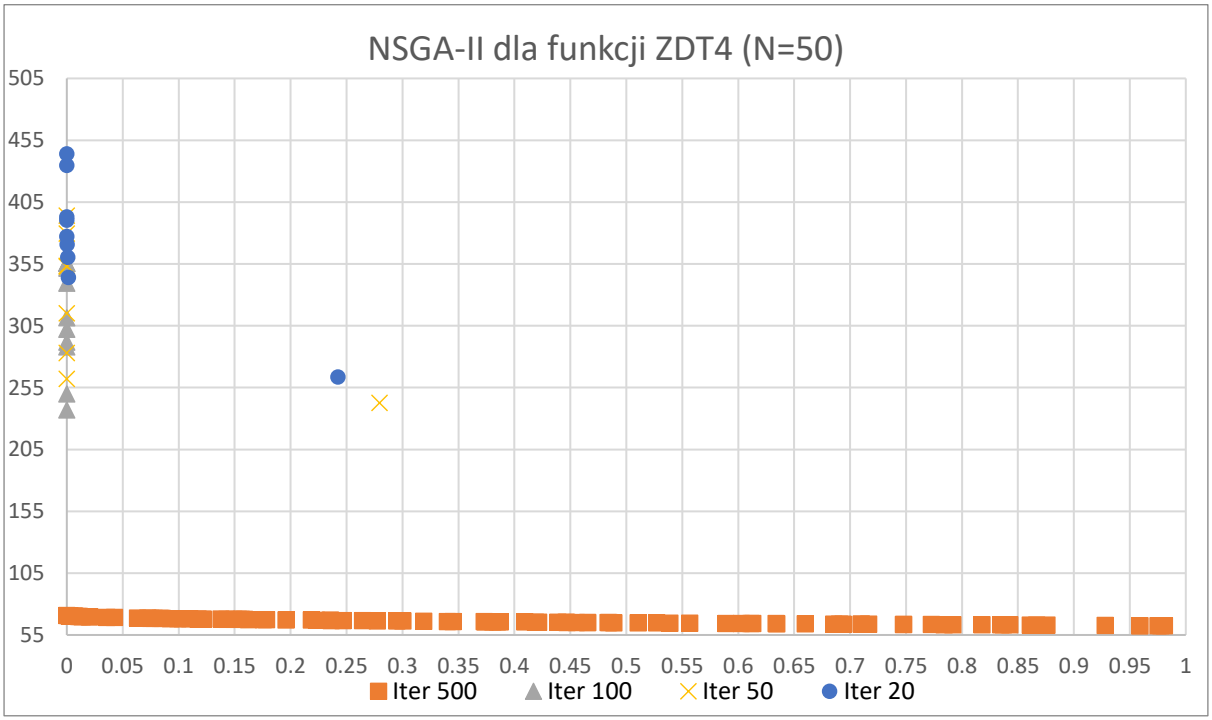

NSGA-II dla funkcji ZDT3 (N=10)
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9
Iter 500
Iter 100
Iter 50
Iter 20

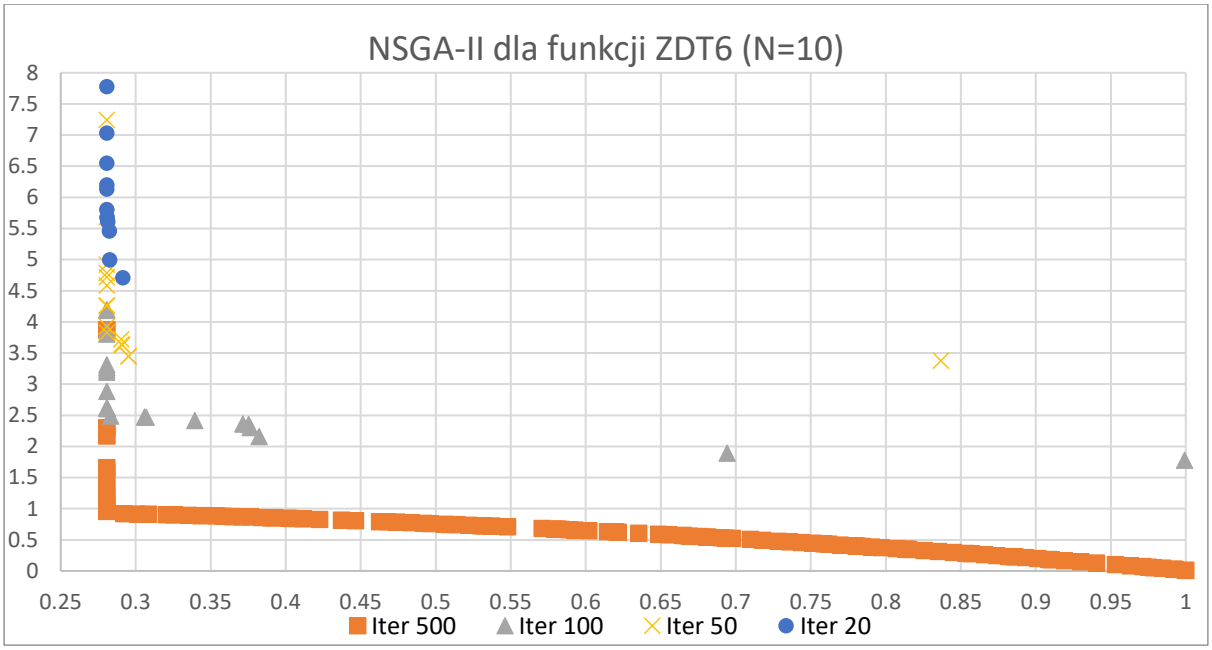

NSGA-II dla funkcji ZDT3 (N=30)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9
Iter 500
Iter 100
Iter 50
Iter 20

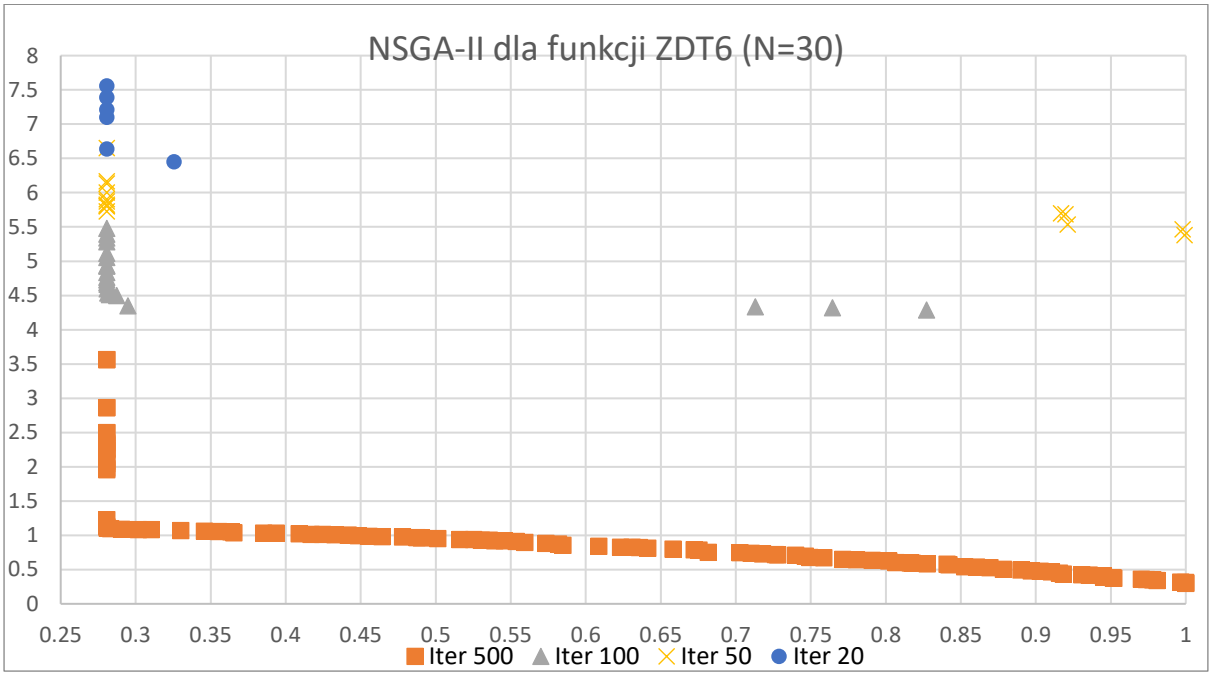

NSGA-II dla funkcji ZDT3 (N=50)
3.7
3.5
3.3
3.1
2.9
2.7
2.5
2.3
2.1
1.9
1.7
1.5
1.3
1.1
0.9
0.7
0.5
0.3
0.1
-0.1
-0.3
-0.5
-0.7
-0.9
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9
Iter 500
Iter 100
Iter 50
Iter 20

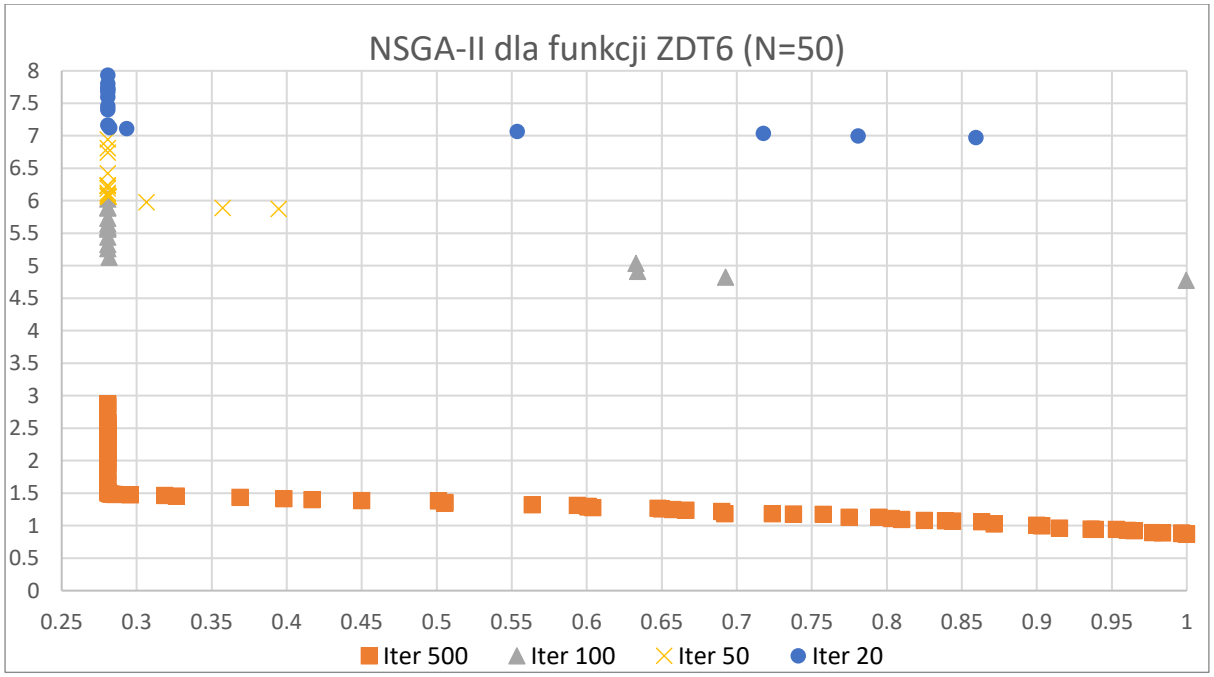

NSGA-II dla funkcji ZDT4 (N=10)
50
47
44
41
38
35
32
29
26
23
20
17
14
11
8
5
2
-1
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9 0.95 1
Iter 500
Iter 100
Iter 50
Iter 20


NSGA-II dla funkcji ZDT4 (N=30)
200
180
160
140
120
100
80
60
40
20
0
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9 0.95 1
Iter 500
Iter 100
Iter 50
Iter 20

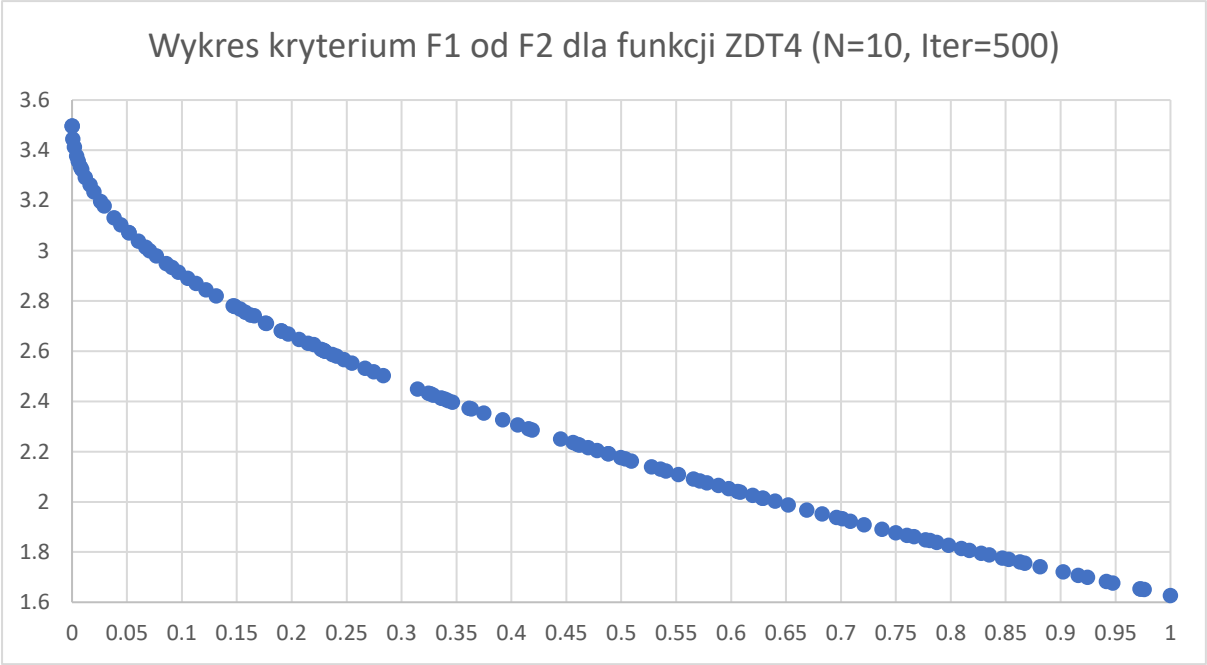

NSGA-II dla funkcji ZDT4 (N=50)
505
455
405
355
305
255
205
155
105
55
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9 0.95 1
Iter 500
Iter 100
Iter 50
Iter 20

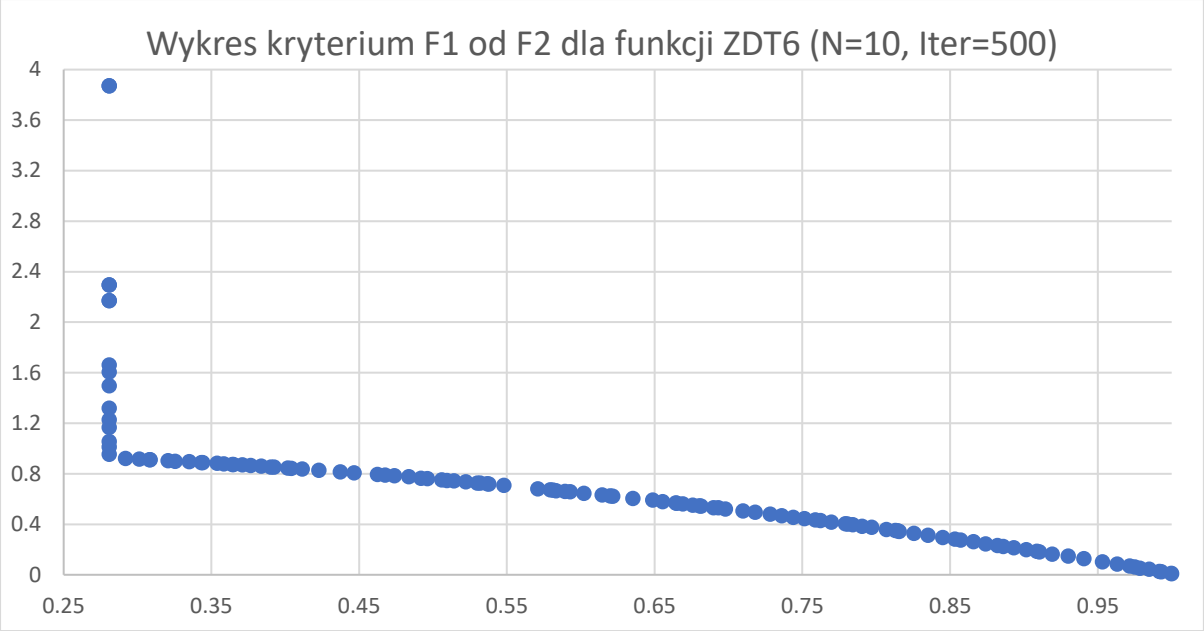

NSGA-II dla funkcji ZDT6 (N=10)
8
7.5
7
6.5
6
5.5
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
0.25
0.3
0.35
0.4
0.45
0.5
0.55
0.6
0.65
0.7
0.75
0.8
0.85
0.9
0.95
1
Iter 500
Iter 100
Iter 50
Iter 20


NSGA-II dla funkcji ZDT6 (N=30)
8
7.5
7
6.5
6
5.5
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
0.25
0.3
0.35
0.4
0.45
0.5
0.55
0.6
0.65
0.7
0.75
0.8
0.85
0.9
0.95
1
Iter 500
Iter 100
Iter 50
Iter 20



Dodatkowo podaje wykresy F1 od F2 dla funkcji ZDT4 i ZDT6, gdyż przy zbyt wysokich wynikach F2 we wczesnych iteracjach krzywe potrafią się wypłaszczyć, więc poniżej przedstawiam otrzymane wykresy krzywych dla 500 iteracji i N=30:




Wykres kryterium F1 od F2 dla funkcji ZDT6 (N=10, Iter=500)
4
3.6
3.2
2.8
2.4
2
1.6
1.2
0.8
0.4
0
0.25
0.35
0.45
0.55
0.65
0.75
0.85
0.95

a